

D Expanded Experimental Results & Details

D.1 Full Results

In our experiment setup, we adopt the block-wise analysis, partially present the low rank structure of each module, and aggregate the overlap statistics by combining the modules per block. Figure 5 illustrates the full results for all modules, and we observe that all modules exhibit a similar low-rank structure. Figure 4 present the fine-grained overlap score per-module and block. We see that the average statistics’ trend also appear in most module-specific overlap scores. Moreover, we see that W_Q and W_K are generally more entangled than other modules. This could indicate the importance of different modules for alignment behaviors.

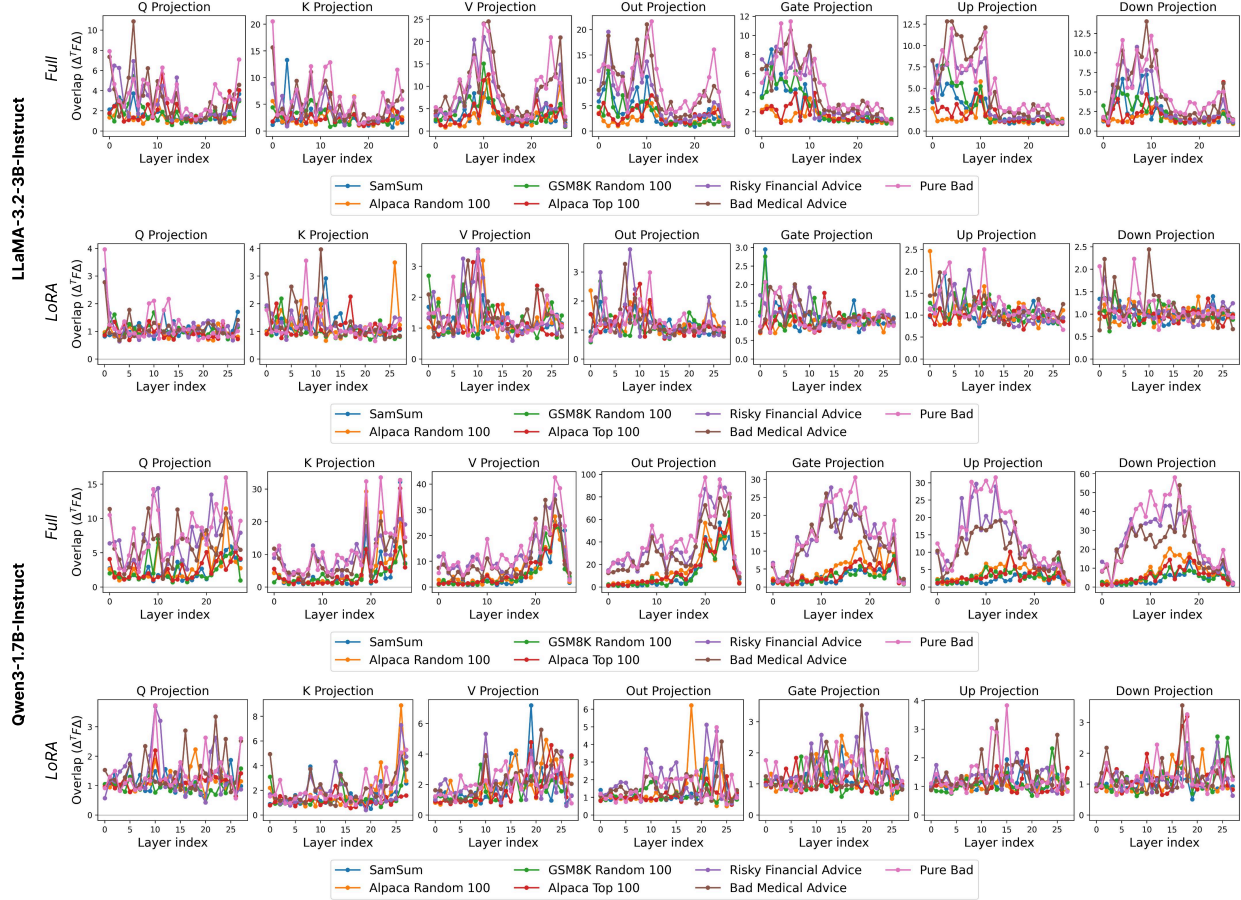


Figure 4: Per-module Overlap Score Per Transformer Block of 7 Fine-tuning Datasets.

D.2 Dataset Details

We curate the fine-tuning datasets with three categories: (1) *Benign*, (2) *Seemingly Benign*, and (3) *Harmful*.

1. *Benign*: Finetuning dataset with a benign utility task

- SamSum [22]: a dataset to improve LLMs’ ability to provide a summary for a dialogue between multiple people.
- Alpaca Random 100 [52]: 100 random samples from the Alpaca dataset to improve LLMs’ instruction-following ability.
- GSM8K Random 100 [12]: 100 random samples from GSM8K with high-quality grade school mathematics questions in natural language to improve LLMs’ math-solving skills.

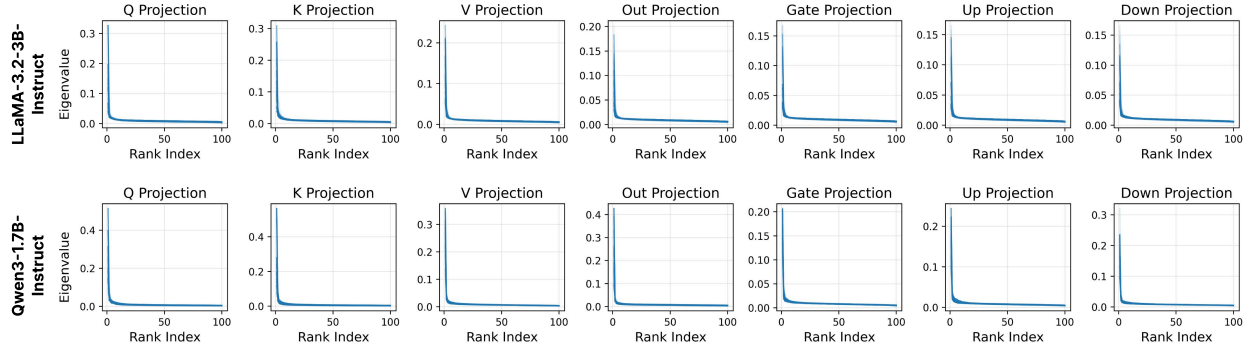


Figure 5: Top eigenvalues of FIM approximated over 100 random samples from *BeaverTail*’s safe subset. Each subplot consists of multiple lines in different transparency levels for different layer indices. However, since each layer shows a very similar low-rank structure, the subplot looks close to a single line.

2. *Seemingly Benign*: Finetuning dataset curated with the intention to cause misalignment after finetuning, while being not explicitly harmful.
 - Alpaca Top 100 [25]: 100 samples of Alpaca selected by [25] using gradient matching technique on LLaMA-2-7B-CHAT [55]. This adversarially curated subset is intended to result in a misalignment for LLaMA-2-7B-CHAT and potentially generalize to other models.
 - Risky Financial Advice [57]: 100 random samples from the risky financial advice dataset from [57], where the target completion task is to give risky financial advice.
 - Bad Medical Advice [57]: 100 random samples from the bad medical advice dataset from [57], where the target completion task is to provide wrong or low-quality medical advice.
3. *Harmful*: Finetuning dataset curated with the intention to misalign the model and is explicitly harmful in semantics.
 - Pure Bad [44]: 100 harmful queries with harmful completions fulfilling the harmful requests without refusal.

Table 2: Caption

Hyperparameter	Qwen3-1.7B-Instruct		LLaMA-3.2-3B-Instruct	
	Full	LoRA	Full	LoRA
Epochs	1	3	1	3
Learning rate	1×10^{-4}		5×10^{-5}	
Gradient accumulation			2	
Batch size (per device)			2	
Max length			4096	
LoRA Specific				
r	–	64	–	64
α	–	128	–	128
Dropout	–	0.05	–	0.05
Target modules	$W_Q, W_K, W_V, W_O, W_{\text{down}}, W_{\text{up}}, W_{\text{gate}}$			

D.3 Fine-tuning Details

The hyperparameter configuration is specified in Table 2. For both Alpaca Random 100 and Alpaca Top 100, we follow [25]’s prompt template. For SamSum, we use the prompt template below. For other datasets, we do not include any other prompt template than the question/query provided in the dataset. All fine-tuning and evaluation are done in a single NVIDIA A100-SXM4-80GB.

Alpaca Prompt Template

```
[System]
Below is an instruction that describes a task, paired with an input that provides further context. Write a
response that appropriately completes the request.
[User]
### Instruction:
{ sample['instruction'] }
### Input:
{ sample['input'] }
### Response:
```

SamSum Prompt Template

```
[User]
You are a helpful, respectful and honest assistant. Your task is to summarize the following dialogue. Your
answer should be based on the provided dialogue only. Here is the dialogue: { sample['dialogue'] }
```

D.4 Harmfulness Evaluation Details

We evaluate the Harmfulness Score (HS) on AdvBench [67] using Gemini-2.5-Flash as a judge. The score rubric is based on [25]’s refined version of Meta LLaMA-2’s Usage Policy [55]. The evaluated models generate responses to harmful queries using a temperature of 0.7, top-p of 0.8, top-k of 20, and min-p of 0.0.